

Homework 7

Problem 1. Let $x \in \mathbb{R}$.

(a) We have $\cos(x + \pi) = -\cos(x)$ and $\sin(x + \pi) = -\sin(x)$. In particular we have $\cos(x + 2\pi) = \cos(x)$ and $\sin(x + 2\pi) = \sin(x)$.

(b) We have $\sin(x) = 0$ if and only if $x = k\pi$ where $k \in \mathbb{Z}$.

(c) We have $\cos(x) = 0$ if and only if $x = k\pi + \frac{1}{2}$ where $k \in \mathbb{Z}$

Proof of (a). By Theorem 4.7.2 (d) we have $\cos(x + \pi) = \cos(x)\cos(\pi) - \sin(x)\sin(\pi)$. By definition of π , we have that $\cos(\pi) = -1$ and $\sin(\pi) = 0$. So we have $\cos(x + \pi) = -\cos(x)$. Similarly, $\sin(x + \pi) = \sin(x)\cos(\pi) + \cos(x)\sin(\pi) = -\sin(x)$. From these results we have, $\cos(x + 2\pi) = \cos((x + \pi) + \pi) = \cos(x + \pi)\cos(\pi) - \sin(x + \pi)\sin(\pi) = \cos(x)$. Symmetrically, $\sin(x + 2\pi) = \sin((x + \pi) + \pi) = \sin(x + \pi)\cos(\pi) + \cos(x + \pi)\sin(\pi) = \sin(x)$. $\square$

Proof of (b). $\Rightarrow$. Suppose $\sin(x) = 0$. For any x we can write $x = k\pi + r$ where $k \in \mathbb{Z}$ and $0 \leq r < \pi$. Then we have that $\sin(x) = \sin(k\pi + r) = 0$. We claim that $\sin(r) = 0$. By properties of π , observe that $\sin(x + \pi) = -\sin(x)$, $\forall x$. In other words, we have that $\sin(x) = 0 \iff \sin(x + \pi) = 0$. So for $\sin(k\pi + r)$ we can repeatedly shift by π k -times toward 0 to get that $\sin(r) = 0$. Suppose for the sake of contradiction that $r \neq 0$. Then by assumption $r \in (0, \pi)$, i.e., $r \in \{x > 0 \mid \sin(x) = 0\}$ and $r < \pi$. But this is a contradiction since by definition $\pi := \inf\{x > 0 \mid \sin(x) = 0\}$. So we must have that $r = 0$.

Thus $x = k\pi$ where $k \in \mathbb{Z}$.

$\Leftarrow$. Suppose $x = k\pi$ where $k \in \mathbb{Z}$. Then we have $\sin(x) = \sin(k\pi)$. Note that we can write $\sin(k\pi) = \sin(0 + k\pi)$. Since $\sin(0) = 0$, then by the same logic that was shown above, $\sin(k\pi) = 0$. $\square$

Proof of (c). $\Rightarrow$. Suppose $\cos(x) = 0$. We first show that $\cos(x) = \sin(x + \frac{\pi}{2})$. Note by the addition property we have

$$\sin\left(x + \frac{\pi}{2}\right) = \sin(x)\cos\left(\frac{\pi}{2}\right) + \cos(x)\sin\left(\frac{\pi}{2}\right). \quad (1)$$

So we need to find the values for $\sin(\frac{\pi}{2})$ and $\cos(\frac{\pi}{2})$. We also have

$$0 = \sin(\pi) = \sin\left(\frac{\pi}{2} + \frac{\pi}{2}\right) = 2\sin\left(\frac{\pi}{2}\right)\cos\left(\frac{\pi}{2}\right). \quad (2)$$

Since $\sin(x) > 0$ for $x \in (0, \pi)$, then $\sin(\frac{\pi}{2}) > 0$. Then by (2), we have $\cos(\frac{\pi}{2}) = 0$. By the addition property for cos we have,

$$-1 = \cos(\pi) = \cos\left(\frac{\pi}{2} + \frac{\pi}{2}\right) = \cos^2\left(\frac{\pi}{2}\right) - \sin^2\left(\frac{\pi}{2}\right). \quad (3)$$

Since $\cos(\frac{\pi}{2}) = 0$, then by (3) we have $1 = \sin^2(\frac{\pi}{2})$. Since we have that $\sin(\frac{\pi}{2}) > 0$, then we have $\sin(\frac{\pi}{2}) = 1$. Then by (1), we have $\sin(x + \frac{\pi}{2}) = \sin(x) \cdot 0 + \cos(x) \cdot 1 = \cos(x)$. So by assumption $\sin(x + \frac{\pi}{2}) = 0$. By (b), $x + \frac{\pi}{2} = k\pi \Rightarrow x = k\pi - \frac{\pi}{2} \Rightarrow x = (k - 1)\pi + \frac{\pi}{2}$. Since $k - 1 \in \mathbb{Z}$ we are done.

$\Leftarrow$. Suppose $x = (k + \frac{1}{2})\pi$ for some $k \in \mathbb{Z}$. Then we have $\cos(k\pi + \frac{\pi}{2}) = \cos((k+1)\pi - \frac{\pi}{2}) = \sin((k+1)\pi) \stackrel{(b)}{=} 0$. $\square$

Problem 2. Let $\tan : (-\frac{\pi}{2}, \frac{\pi}{2}) \rightarrow \mathbb{R}$ such that $\tan(x) := \frac{\sin(x)}{\cos(x)}$. Show that $\tan$ is differentiable and monotone increasing, with $(\tan(x))' = 1 + \tan^2(x)$, and $\lim_{x \rightarrow \frac{\pi}{2}} \tan(x) = +\infty$ and $\lim_{x \rightarrow -\frac{\pi}{2}} \tan(x) = -\infty$. Conclude that $\tan$ is in fact a bijection from $(-\frac{\pi}{2}, \frac{\pi}{2}) \rightarrow \mathbb{R}$, and thus has an inverse function $\tan^{-1} : \mathbb{R} \rightarrow (-\frac{\pi}{2}, \frac{\pi}{2})$. Show that $(\tan^{-1}(x))' = \frac{1}{1+x^2}$.

Proof. Note that $\sin, \cos$ are both differentiable function on $\mathbb{R}$ and in particular $(-\frac{\pi}{2}, \frac{\pi}{2})$. Also note that $\cos(x) = 0 \iff x = k\pi + \frac{\pi}{2}$ and for any $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$, $x \neq k\pi + \frac{\pi}{2}$, $\forall k$. Hence we have that $\cos$ is non zero on $(-\frac{\pi}{2}, \frac{\pi}{2})$. This means that $\tan(x)$ is differentiable. We have

$$\frac{d}{dx} \tan(x) = \frac{d}{dx} \frac{\sin(x)}{\cos(x)} = \frac{\cos(x)\cos(x) - \sin(x)(-\sin(x))}{(\cos(x))^2} = 1 + \tan^2(x).$$

Note that $1 + \tan^2(x) > 0$, $\forall x \in (-\frac{\pi}{2}, \frac{\pi}{2})$. So $\tan(x)$ is a strictly increasing function on its domain. Now we claim that $\lim_{x \rightarrow \frac{\pi}{2}} \tan(x) = +\infty$ and $\lim_{x \rightarrow -\frac{\pi}{2}} \tan(x) = -\infty$. Indeed, as $x \rightarrow \frac{\pi}{2}$ we have that $\sin(x) \rightarrow 1$ and $\cos(x) \rightarrow 0^+$ since we must approach $\frac{\pi}{2}$ from the left to stay on the domain of $\tan$. This means that $\tan(x) := \frac{\sin(x)}{\cos(x)}$ gets arbitrarily positive as $x \rightarrow \frac{\pi}{2}$, i.e., $\tan(x) \rightarrow +\infty$. Similarly as $x \rightarrow -\frac{\pi}{2}$, $\sin(x) \rightarrow -1$ and $\cos(x) \rightarrow 0^+$. So $\frac{\sin(x)}{\cos(x)}$ gets arbitrarily negative, i.e., $\tan(x) \rightarrow -\infty$. Since $\tan$ is monotone increasing, $\tan$ is injective. Also since $\tan$ is differentiable, and in particular continuous, we have that $\tan$ is surjective since on its domain it approaches $-\infty$ and $+\infty$, i.e., by IVT $\tan$ must reach all values in $\mathbb{R}$. It follows that $\tan$ is bijective and hence an inverse function $\tan^{-1} : \mathbb{R} \rightarrow (-\frac{\pi}{2}, \frac{\pi}{2})$ exists. By the inverse function theorem since $\tan'(x) = 1 + \tan^2(x) > 0$, we have,

$$(\tan^{-1}(x))' = \frac{1}{\tan'(\tan^{-1}(x))} = \frac{1}{1 + \tan^2(\tan^{-1}(x))} = \frac{1}{1 + x^2}.$$

$\square$

Problem 3. (Basic properties of $C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$)

- (a) (Boundedness) If $f \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$, then f is bounded.
- (b) (Vector space and algebra properties) If $f, g \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$, then the functions $f + g$, $f - g$, and fg are also in $C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$. Also if $c \in \mathbb{C}$, then the function cf is also in $C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$.
- (c) (Closure under uniform limits) If $(f_n)_{n=1}^\infty$ is a sequence of functions in $C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$ which converges uniformly to another function $f : \mathbb{R} \rightarrow \mathbb{C}$, then f is also in $C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$.

Proof of (a). We first show that f is bounded on $[0, 1]$. Note that $[0, 1]$ is a compact interval in $\mathbb{R}$. Since by assumption f is continuous, then the image of f on $[0, 1]$ is also compact, and in particular bounded. In other words we have that there exists $M > 0$ such that for any $x \in [0, 1]$, $|f(x)| < M$. Let $x \in \mathbb{R}$. Note that x can be written in terms of its integer and fractional part as $x = k + y$, where $k \in \mathbb{Z}$

and $y \in [0, 1)$. Since f is $\mathbb{Z}$ -periodic by assumption, then we have $|f(x)| = |f(k + y)| = |f(y)| < M$.

Thus f is bounded. $\square$

Proof of (b). Let $f, g \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$. Then for any $x \in \mathbb{R}$ and $k \in \mathbb{Z}$, we have that $f(x + k) = f(x)$ and $g(x + k) = g(x)$. Since f, g both continuous, then $f + g$ is also continuous by the additive limit law. We also have that since $\mathbb{C}$ is a vector space, then $(f + g) : \mathbb{R} \rightarrow \mathbb{C}$. It suffices to show that $f + g$ is $\mathbb{Z}$ -periodic. Take any $x \in \mathbb{R}, k \in \mathbb{Z}$. Then $(f + g)(x + k) = f(x + k) + g(x + k) = f(x) + g(x) = (f + g)(x)$. Thus $f + g$ is $\mathbb{Z}$ -periodic. The proof for $f - g$ is symmetric. Continuity of fg follows from the multiplication limit law, and since $\mathbb{C}$ is a vector space, then $fg : \mathbb{R} \rightarrow \mathbb{C}$. Take any $x \in \mathbb{R}, k \in \mathbb{Z}$. Then $(fg)(x + k) = f(x + k)g(x + k) = f(x)g(x) = (fg)(x)$. So fg is $\mathbb{Z}$ -periodic. Continuity of cf follows from scalar multiplication of limits. The fact that $cf : \mathbb{R} \rightarrow \mathbb{C}$ follows from the same logic above. Take any $x \in \mathbb{R}, k \in \mathbb{Z}$. Then $(cf)(x + k) = cf(x + k) = cf(x) = (cf)(x)$. $\square$

Proof of (c). Let $(f_n)_{n=1}^\infty$ be a sequence of functions in $C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$ which converges uniformly to another function $f : \mathbb{R} \rightarrow \mathbb{C}$. Let $\varepsilon > 0$ uniform convergence there exists an N such that for all $x \in \mathbb{R}$ and $n \geq N$, we have $|f_n(x) - f(x)| < \varepsilon/2$. Let $k \in \mathbb{Z}$. Fix $n > N$. Then for all $x \in \mathbb{R}$, we have $|f(x) - f(x + k)| \leq |f(x) - f_n(x)| + |f_n(x) - f_n(x + k)| + |f_n(x + k) - f(x + k)| < \varepsilon/2 + 0 + \varepsilon/2 = \varepsilon$. Since ε was arbitrary, then we have that $f(x) = f(x + k)$, i.e., f is $\mathbb{Z}$ -periodic. Also note that since $f_n \rightarrow f$ uniformly and each f_n continuous, then f is also continuous. It follows that $f \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$. $\square$

Problem 4. Show that $C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$ with the sup norm metric d_∞ is a metric space. Furthermore show that this metric space is complete.

Proof. We need to verify the four properties of metric spaces.

- (1) Take any $f, g \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$ such that $f(x) \neq g(x)$ for at least one $x \in \mathbb{R}$. Then $d_\infty(f, g) = \sup_{x \in \mathbb{R}} |f(x) - g(x)| > 0$ since there exists $x \in \mathbb{R}$ where $f(x) \neq g(x)$, i.e., $|f(x) - g(x)| > 0$.
- (2) Take $f \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$. Then $d_\infty(f, f) = \sup_{x \in \mathbb{R}} |f(x) - f(x)| = 0$.
- (3) Take any $f, g \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$. Then $d_\infty(f, g) = \sup_{x \in \mathbb{R}} |f(x) - g(x)| = \sup_{x \in \mathbb{R}} |g(x) - f(x)| = d_\infty(g, f)$.
- (4) Take any $f, g, h \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$. Then we have,

$$\begin{aligned}
 d_\infty(f, g) &= \sup_{x \in \mathbb{R}} |f(x) - g(x)| \\
 &\triangleq \sup_{x \in \mathbb{R}} (|f(x) - h(x)| + |h(x) - g(x)|) \\
 &\leq \sup_{x \in \mathbb{R}} |f(x) - h(x)| + \sup_{x \in \mathbb{R}} |h(x) - g(x)| \\
 &= d_\infty(f, h) + d_\infty(h, g)
 \end{aligned}$$

We now show that $(C(\mathbb{R}/\mathbb{Z}; \mathbb{C}), d_\infty)$ is complete. Since all functions in $C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$ are continuous and bounded, then $C(\mathbb{R}/\mathbb{Z}; \mathbb{C}) \subset C(\mathbb{R} \rightarrow \mathbb{C})$. Since $\mathbb{C}$ is complete when equipped with its standard metric,

then $C(\mathbb{R} \rightarrow \mathbb{C})$ is also complete when equipped with the sup norm metric by Theorem 3.4.5. Note by Lemma 5.1.5 (c), we have that $C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$ is closed with respect to the sup norm metric. Then for any cauchy sequence $(f_n)_n \subset C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$ we have that $(f_n)_n \subset C(\mathbb{R} \rightarrow \mathbb{C})$. Since $C(\mathbb{R} \rightarrow \mathbb{C})$ cauchy, then f_n converges in $C(\mathbb{R} \rightarrow \mathbb{C})$. But since $C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$ is closed and $f_n \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$ for all n , then we must have that f_n converges in $C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$. Hence $C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$ is complete with respect to the sup norm metric. □